

**5203ef7ca5d4ee5719268524a
c7984026c0b04d1**

Instructions

- Following Playwright test failed.
- Explain why, be concise, respect Playwright best practices.
- Provide a snippet of code with the fix, if possible.

Test info

- Name: milosvasic-ru-a11y.spec.js >> milosvasic.ru — accessibility >> axe-core accessibility scan — product (no critical/serious violations)
- Location: tests/milosvasic-ru-a11y.spec.js:20:5

Error details

Error: product must have no critical/serious ally violations

`expect(received).toEqual(expected) // deep equality`

- Expected - 1
+ Received + 513

- Array []
+ Array [
+ Object {
+ "description": "Ensure the contrast between foreground and background colors meets WCAG 2 AA minimum contrast ratio thresholds",
+ "help": "Elements must meet minimum color contrast ratio thresholds",
+ "helpUrl": "https://dequeuniversity.com/rules/axe/4.11/color-contrast?application=playwright",
+ "id": "color-contrast",

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+     "impact": "serious",
+     "nodes": Array [
+       Object {
+         "all": Array [],
+         "any": Array [
+           Object {
+             "data": Object {
+               "bgColor": "#e9ecef",
+               "contrastRatio": 3.95,
+               "expectedContrastRatio": "4.5:1",
+               "fgColor": "#6c757d",
+               "fontSize": "9.9pt (13.2px)",
+               "fontWeight": "normal",
+               "messageKey": null,
+             },
+             "id": "color-contrast",
+             "impact": "serious",
+             "message": "Element has insufficient color contrast
+ of 3.95 (foreground color: #6c757d, background color: #e9ecef,
+ font size: 9.9pt (13.2px), font weight: normal). Expected contrast
+ ratio of 4.5:1",
+             "relatedNodes": Array [
+               Object {
+                 "html": "<span class=\"od-tag--
+ license\">license: TBD</span>",
+                 "target": Array [
+                   ".od-tag--license",
+                 ],
+               },
+             ],
+           },
+         ],
+       },
+       "failureSummary": "Fix any of the following:
+ Element has insufficient color contrast of 3.95 (foreground
+ color: #6c757d, background color: #e9ecef, font size: 9.9pt
+ (13.2px), font weight: normal). Expected contrast ratio of 4.5:1",
+       "html": "<span class=\"od-tag--license\">license: TBD</
+ span>",
+       "impact": "serious",
+       "none": Array [],
+       "target": Array [
+         ".od-tag--license",
+       ],
+     },
+     Object {
+       "all": Array [],
+       "any": Array [
+         Object {
+           "data": Object {
+             "bgColor": "#e9ecef",
+             "contrastRatio": 3.95,
+             "expectedContrastRatio": "4.5:1",

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+         "fgColor": "#6c757d",
+         "fontSize": "9.9pt (13.2px)",
+         "fontWeight": "normal",
+         "messageKey": null,
+     },
+     "id": "color-contrast",
+     "impact": "serious",
+     "message": "Element has insufficient color contrast
of 3.95 (foreground color: #6c757d, background color: #e9ecef,
font size: 9.9pt (13.2px), font weight: normal). Expected contrast
ratio of 4.5:1",
+     "relatedNodes": Array [
+         Object {
+             "html": "<span class=\"od-chip\">Go</span>",
+             "target": Array [
+                 ".od-chip:nth-child(1)",
+             ],
+         },
+     ],
+ },
+ ],
+     "failureSummary": "Fix any of the following:
+ Element has insufficient color contrast of 3.95 (foreground
color: #6c757d, background color: #e9ecef, font size: 9.9pt
(13.2px), font weight: normal). Expected contrast ratio of 4.5:1",
+     "html": "<span class=\"od-chip\">Go</span>",
+     "impact": "serious",
+     "none": Array [],
+     "target": Array [
+         ".od-chip:nth-child(1)",
+     ],
+ },
+ Object {
+     "all": Array [],
+     "any": Array [
+         Object {
+             "data": Object {
+                 "bgColor": "#e9ecef",
+                 "contrastRatio": 3.95,
+                 "expectedContrastRatio": "4.5:1",
+                 "fgColor": "#6c757d",
+                 "fontSize": "9.9pt (13.2px)",
+                 "fontWeight": "normal",
+                 "messageKey": null,
+             },
+             "id": "color-contrast",
+             "impact": "serious",
+             "message": "Element has insufficient color contrast
of 3.95 (foreground color: #6c757d, background color: #e9ecef,
font size: 9.9pt (13.2px), font weight: normal). Expected contrast
ratio of 4.5:1",
+             "relatedNodes": Array [

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+         Object {
+             "html": "<span class=\"od-chip\">Gin</span>",
+             "target": Array [
+                 ".od-chip:nth-child(2)",
+             ],
+         },
+     ],
+ },
+ },
+     "failureSummary": "Fix any of the following:
+ Element has insufficient color contrast of 3.95 (foreground
color: #6c757d, background color: #e9ecef, font size: 9.9pt
(13.2px), font weight: normal). Expected contrast ratio of 4.5:1",
+     "html": "<span class=\"od-chip\">Gin</span>",
+     "impact": "serious",
+     "none": Array [],
+     "target": Array [
+         ".od-chip:nth-child(2)",
+     ],
+ },
+ Object {
+     "all": Array [],
+     "any": Array [
+         Object {
+             "data": Object {
+                 "bgColor": "#e9ecef",
+                 "contrastRatio": 3.95,
+                 "expectedContrastRatio": "4.5:1",
+                 "fgColor": "#6c757d",
+                 "fontSize": "9.9pt (13.2px)",
+                 "fontWeight": "normal",
+                 "messageKey": null,
+             },
+             "id": "color-contrast",
+             "impact": "serious",
+             "message": "Element has insufficient color contrast
of 3.95 (foreground color: #6c757d, background color: #e9ecef,
font size: 9.9pt (13.2px), font weight: normal). Expected contrast
ratio of 4.5:1",
+             "relatedNodes": Array [
+                 Object {
+                     "html": "<span class=\"od-chip\">HTTP/3 QUIC</
span>",
+                     "target": Array [
+                         ".od-chip:nth-child(3)",
+                     ],
+                 },
+             ],
+         },
+     ],
+ },
+     "failureSummary": "Fix any of the following:
+ Element has insufficient color contrast of 3.95 (foreground

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color: #6c757d, background color: #e9ecef, font size: 9.9pt
(13.2px), font weight: normal). Expected contrast ratio of 4.5:1",
+       "html": "<span class=\"od-chip\">HTTP/3 QUIC</span>",
+       "impact": "serious",
+       "none": Array [],
+       "target": Array [
+         ".od-chip:nth-child(3)",
+       ],
+     },
+     Object {
+       "all": Array [],
+       "any": Array [
+         Object {
+           "data": Object {
+             "bgColor": "#e9ecef",
+             "contrastRatio": 3.95,
+             "expectedContrastRatio": "4.5:1",
+             "fgColor": "#6c757d",
+             "fontSize": "9.9pt (13.2px)",
+             "fontWeight": "normal",
+             "messageKey": null,
+           },
+           "id": "color-contrast",
+           "impact": "serious",
+           "message": "Element has insufficient color contrast
of 3.95 (foreground color: #6c757d, background color: #e9ecef,
font size: 9.9pt (13.2px), font weight: normal). Expected contrast
ratio of 4.5:1",
+           "relatedNodes": Array [
+             Object {
+               "html": "<span class=\"od-chip\">PostgreSQL</
span>",
+               "target": Array [
+                 ".od-chip:nth-child(4)",
+               ],
+             },
+           ],
+         },
+       ],
+       "failureSummary": "Fix any of the following:
+     Element has insufficient color contrast of 3.95 (foreground
color: #6c757d, background color: #e9ecef, font size: 9.9pt
(13.2px), font weight: normal). Expected contrast ratio of 4.5:1",
+       "html": "<span class=\"od-chip\">PostgreSQL</span>",
+       "impact": "serious",
+       "none": Array [],
+       "target": Array [
+         ".od-chip:nth-child(4)",
+       ],
+     },
+     Object {
+       "all": Array [],

```

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+         "any": Array [
+             Object {
+                 "data": Object {
+                     "bgColor": "#e9ecef",
+                     "contrastRatio": 3.95,
+                     "expectedContrastRatio": "4.5:1",
+                     "fgColor": "#6c757d",
+                     "fontSize": "9.9pt (13.2px)",
+                     "fontWeight": "normal",
+                     "messageKey": null,
+                 },
+                 "id": "color-contrast",
+                 "impact": "serious",
+                 "message": "Element has insufficient color contrast
of 3.95 (foreground color: #6c757d, background color: #e9ecef,
font size: 9.9pt (13.2px), font weight: normal). Expected contrast
ratio of 4.5:1",
+                 "relatedNodes": Array [
+                     Object {
+                         "html": "<span class=\"od-chip\">SQLite</span>",
+                         "target": Array [
+                             ".od-chip:nth-child(5)",
+                         ],
+                     },
+                 ],
+             },
+         ],
+         "failureSummary": "Fix any of the following:
+ Element has insufficient color contrast of 3.95 (foreground
color: #6c757d, background color: #e9ecef, font size: 9.9pt
(13.2px), font weight: normal). Expected contrast ratio of 4.5:1",
+         "html": "<span class=\"od-chip\">SQLite</span>",
+         "impact": "serious",
+         "none": Array [],
+         "target": Array [
+             ".od-chip:nth-child(5)",
+         ],
+     },
+     Object {
+         "all": Array [],
+         "any": Array [
+             Object {
+                 "data": Object {
+                     "bgColor": "#e9ecef",
+                     "contrastRatio": 3.95,
+                     "expectedContrastRatio": "4.5:1",
+                     "fgColor": "#6c757d",
+                     "fontSize": "9.9pt (13.2px)",
+                     "fontWeight": "normal",
+                     "messageKey": null,
+                 },
+                 "id": "color-contrast",

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+         "impact": "serious",
+         "message": "Element has insufficient color contrast
of 3.95 (foreground color: #6c757d, background color: #e9ecef,
font size: 9.9pt (13.2px), font weight: normal). Expected contrast
ratio of 4.5:1",
+         "relatedNodes": Array [
+             Object {
+                 "html": "<span class=\"od-chip\">SQLCipher</
span>",
+                 "target": Array [
+                     ".od-chip:nth-child(6)",
+                 ],
+             },
+         ],
+     },
+ ],
+ "failureSummary": "Fix any of the following:
+ Element has insufficient color contrast of 3.95 (foreground
color: #6c757d, background color: #e9ecef, font size: 9.9pt
(13.2px), font weight: normal). Expected contrast ratio of 4.5:1",
+     "html": "<span class=\"od-chip\">SQLCipher</span>",
+     "impact": "serious",
+     "none": Array [],
+     "target": Array [
+         ".od-chip:nth-child(6)",
+     ],
+ },
+ Object {
+     "all": Array [],
+     "any": Array [
+         Object {
+             "data": Object {
+                 "bgColor": "#e9ecef",
+                 "contrastRatio": 3.95,
+                 "expectedContrastRatio": "4.5:1",
+                 "fgColor": "#6c757d",
+                 "fontSize": "9.9pt (13.2px)",
+                 "fontWeight": "normal",
+                 "messageKey": null,
+             },
+             "id": "color-contrast",
+             "impact": "serious",
+             "message": "Element has insufficient color contrast
of 3.95 (foreground color: #6c757d, background color: #e9ecef,
font size: 9.9pt (13.2px), font weight: normal). Expected contrast
ratio of 4.5:1",
+             "relatedNodes": Array [
+                 Object {
+                     "html": "<span class=\"od-chip\">Redis</span>",
+                     "target": Array [
+                         ".od-chip:nth-child(7)",
+                     ],
+                 },

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+         },
+     ],
+ },
+ ],
+     "failureSummary": "Fix any of the following:
+   Element has insufficient color contrast of 3.95 (foreground
+   color: #6c757d, background color: #e9ecef, font size: 9.9pt
+   (13.2px), font weight: normal). Expected contrast ratio of 4.5:1",
+     "html": "<span class=\"od-chip\">Redis</span>",
+     "impact": "serious",
+     "none": Array [],
+     "target": Array [
+       ".od-chip:nth-child(7)",
+     ],
+   },
+   Object {
+     "all": Array [],
+     "any": Array [
+       Object {
+         "data": Object {
+           "bgColor": "#e9ecef",
+           "contrastRatio": 3.95,
+           "expectedContrastRatio": "4.5:1",
+           "fgColor": "#6c757d",
+           "fontSize": "9.9pt (13.2px)",
+           "fontWeight": "normal",
+           "messageKey": null,
+         },
+         "id": "color-contrast",
+         "impact": "serious",
+         "message": "Element has insufficient color contrast
+ of 3.95 (foreground color: #6c757d, background color: #e9ecef,
+ font size: 9.9pt (13.2px), font weight: normal). Expected contrast
+ ratio of 4.5:1",
+         "relatedNodes": Array [
+           Object {
+             "html": "<span class=\"od-chip\">Angular 19</
+ span>",
+             "target": Array [
+               ".od-chip:nth-child(8)",
+             ],
+           },
+         ],
+       },
+     ],
+   },
+     "failureSummary": "Fix any of the following:
+   Element has insufficient color contrast of 3.95 (foreground
+   color: #6c757d, background color: #e9ecef, font size: 9.9pt
+   (13.2px), font weight: normal). Expected contrast ratio of 4.5:1",
+     "html": "<span class=\"od-chip\">Angular 19</span>",
+     "impact": "serious",
+     "none": Array [],

```



```

+         "target": Array [
+             ".od-chip:nth-child(8)",
+         ],
+     },
+     Object {
+         "all": Array [],
+         "any": Array [
+             Object {
+                 "data": Object {
+                     "bgColor": "#e9ecef",
+                     "contrastRatio": 3.95,
+                     "expectedContrastRatio": "4.5:1",
+                     "fgColor": "#6c757d",
+                     "fontSize": "9.9pt (13.2px)",
+                     "fontWeight": "normal",
+                     "messageKey": null,
+                 },
+                 "id": "color-contrast",
+                 "impact": "serious",
+                 "message": "Element has insufficient color contrast
+ of 3.95 (foreground color: #6c757d, background color: #e9ecef,
+ font size: 9.9pt (13.2px), font weight: normal). Expected contrast
+ ratio of 4.5:1",
+                 "relatedNodes": Array [
+                     Object {
+                         "html": "<span class=\"od-chip\">Tauri 2.0</
+ span>",
+                         "target": Array [
+                             ".od-chip:nth-child(9)",
+                         ],
+                     },
+                 ],
+             },
+         ],
+     },
+     "failureSummary": "Fix any of the following:
+ Element has insufficient color contrast of 3.95 (foreground
+ color: #6c757d, background color: #e9ecef, font size: 9.9pt
+ (13.2px), font weight: normal). Expected contrast ratio of 4.5:1",
+     "html": "<span class=\"od-chip\">Tauri 2.0</span>",
+     "impact": "serious",
+     "none": Array [],
+     "target": Array [
+         ".od-chip:nth-child(9)",
+     ],
+ },
+ Object {
+     "all": Array [],
+     "any": Array [
+         Object {
+             "data": Object {
+                 "bgColor": "#e9ecef",
+                 "contrastRatio": 3.95,

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+         "expectedContrastRatio": "4.5:1",
+         "fgColor": "#6c757d",
+         "fontSize": "9.9pt (13.2px)",
+         "fontWeight": "normal",
+         "messageKey": null,
+     },
+     "id": "color-contrast",
+     "impact": "serious",
+     "message": "Element has insufficient color contrast
of 3.95 (foreground color: #6c757d, background color: #e9ecef,
font size: 9.9pt (13.2px), font weight: normal). Expected contrast
ratio of 4.5:1",
+     "relatedNodes": Array [
+         Object {
+             "html": "<span class=\"od-chip\">Kotlin</span>",
+             "target": Array [
+                 ".od-chip:nth-child(10)",
+             ],
+         },
+     ],
+ },
+ ],
+ },
+ },
+     "failureSummary": "Fix any of the following:
+   Element has insufficient color contrast of 3.95 (foreground
color: #6c757d, background color: #e9ecef, font size: 9.9pt
(13.2px), font weight: normal). Expected contrast ratio of 4.5:1",
+     "html": "<span class=\"od-chip\">Kotlin</span>",
+     "impact": "serious",
+     "none": Array [],
+     "target": Array [
+         ".od-chip:nth-child(10)",
+     ],
+ },
+ },
+ Object {
+     "all": Array [],
+     "any": Array [
+         Object {
+             "data": Object {
+                 "bgColor": "#e9ecef",
+                 "contrastRatio": 3.95,
+                 "expectedContrastRatio": "4.5:1",
+                 "fgColor": "#6c757d",
+                 "fontSize": "9.9pt (13.2px)",
+                 "fontWeight": "normal",
+                 "messageKey": null,
+             },
+             "id": "color-contrast",
+             "impact": "serious",
+             "message": "Element has insufficient color contrast
of 3.95 (foreground color: #6c757d, background color: #e9ecef,
font size: 9.9pt (13.2px), font weight: normal). Expected contrast
ratio of 4.5:1",

```

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+         "relatedNodes": Array [
+             Object {
+                 "html": "<span class=\"od-chip\">Swift</span>",
+                 "target": Array [
+                     ".od-chip:nth-child(11)",
+                 ],
+             },
+         ],
+     },
+ ],
+ },
+ "failureSummary": "Fix any of the following:
+ Element has insufficient color contrast of 3.95 (foreground
+ color: #6c757d, background color: #e9ecef, font size: 9.9pt
+ (13.2px), font weight: normal). Expected contrast ratio of 4.5:1",
+     "html": "<span class=\"od-chip\">Swift</span>",
+     "impact": "serious",
+     "none": Array [],
+     "target": Array [
+         ".od-chip:nth-child(11)",
+     ],
+ },
+ Object {
+     "all": Array [],
+     "any": Array [
+         Object {
+             "data": Object {
+                 "bgColor": "#e9ecef",
+                 "contrastRatio": 3.95,
+                 "expectedContrastRatio": "4.5:1",
+                 "fgColor": "#6c757d",
+                 "fontSize": "9.9pt (13.2px)",
+                 "fontWeight": "normal",
+                 "messageKey": null,
+             },
+             "id": "color-contrast",
+             "impact": "serious",
+             "message": "Element has insufficient color contrast
+ of 3.95 (foreground color: #6c757d, background color: #e9ecef,
+ font size: 9.9pt (13.2px), font weight: normal). Expected contrast
+ ratio of 4.5:1",
+             "relatedNodes": Array [
+                 Object {
+                     "html": "<span class=\"od-chip\">Docker</span>",
+                     "target": Array [
+                         ".od-chip:nth-child(12)",
+                     ],
+                 },
+             ],
+         },
+     ],
+ },
+ ],
+ },
+ "failureSummary": "Fix any of the following:
+ Element has insufficient color contrast of 3.95 (foreground

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```

color: #6c757d, background color: #e9ecef, font size: 9.9pt
(13.2px), font weight: normal). Expected contrast ratio of 4.5:1",
+       "html": "<span class=\"od-chip\">Dock</span>",
+       "impact": "serious",
+       "none": Array [],
+       "target": Array [
+         ".od-chip:nth-child(12)",
+       ],
+     },
+     Object {
+       "all": Array [],
+       "any": Array [
+         Object {
+           "data": Object {
+             "bgColor": "#f8f9fa",
+             "contrastRatio": 4.44,
+             "expectedContrastRatio": "4.5:1",
+             "fgColor": "#6c757d",
+             "fontSize": "9.9pt (13.2px)",
+             "fontWeight": "normal",
+             "messageKey": null,
+           },
+           "id": "color-contrast",
+           "impact": "serious",
+           "message": "Element has insufficient color contrast
of 4.44 (foreground color: #6c757d, background color: #f8f9fa,
font size: 9.9pt (13.2px), font weight: normal). Expected contrast
ratio of 4.5:1",
+           "relatedNodes": Array [
+             Object {
+               "html": "<figure class=\"od-diagram\" data-
slug=\"helixtrack\">",
+               "target": Array [
+                 "figure",
+               ],
+             },
+           ],
+         },
+       ],
+       "failureSummary": "Fix any of the following:
+ Element has insufficient color contrast of 4.44 (foreground
color: #6c757d, background color: #f8f9fa, font size: 9.9pt
(13.2px), font weight: normal). Expected contrast ratio of 4.5:1",
+       "html": "<figcaption class=\"od-section__eyebrow\"
style=\"margin-top:var(--od-space-2)\">// architecture</
figcaption>",
+       "impact": "serious",
+       "none": Array [],
+       "target": Array [
+         "figcaption",
+       ],
+     },
+   ],

```

```
+ ],
+ "tags": Array [
+   "cat.color",
+   "wcag2aa",
+   "wcag143",
+   "TTv5",
+   "TT13.c",
+   "EN-301-549",
+   "EN-9.1.4.3",
+   "ACT",
+   "RGAAv4",
+   "RGAA-3.2.1",
+ ],
+ },
+ ]
```

Page snapshot

- generic [active] [ref=e1]:
 - link "Skip to content" [ref=e2]:
 - /url: "#main"
 - banner [ref=e3]:
 - link "Miloš Vasić" [ref=e4]:
 - /url: ../index.html
 - generic [ref=e5]:
 - navigation "Primary" [ref=e6]:
 - link "Home" [ref=e7]:
 - /url: ../index.html
 - link "Products" [ref=e8]:
 - /url: ../index.html#products
 - button "Toggle light or dark theme" [ref=e9]
 - [cursor=pointer]:
 - img [ref=e10]
 - main [ref=e12]:
 - article [ref=e13]:
 - paragraph [ref=e14]: "// tier: helix-primary · order 1"
 - 'heading "HelixTrack beta license: TBD" [level=1] [ref=e15]':
 - text: HelixTrack
 - generic [ref=e16]:
 - generic [ref=e17]: beta
 - generic [ref=e18]: "license: TBD"
 - generic [ref=e19]:
 - generic [ref=e20]: Go
 - generic [ref=e21]: Gin
 - generic [ref=e22]: HTTP/3 QUIC
 - generic [ref=e23]: PostgreSQL
 - generic [ref=e24]: SQLite
 - generic [ref=e25]: SQLCipher

- generic [ref=e26]: Redis
- generic [ref=e27]: Angular 19
- generic [ref=e28]: Tauri 2.0
- generic [ref=e29]: Kotlin
- generic [ref=e30]: Swift
- generic [ref=e31]: Docker
- heading "Source" [level=2] [ref=e32]
- list [ref=e33]:
 - listitem [ref=e34]:
 - link "Helix-Track/Core" [ref=e35]:
 - /url: <https://github.com/Helix-Track/Core>
 - text: Helix-Track/Core
 - img [ref=e36]
 - listitem [ref=e38]:
 - link "Helix-Track/Website" [ref=e39]:
 - /url: <https://github.com/Helix-Track/Website>
 - text: Helix-Track/Website
 - img [ref=e40]
- figure "// architecture" [ref=e42]:
 - img [ref=e43]:
 - generic [ref=e45]: HelixTrack – architecture map
 - generic [ref=e46]: Native clients
 - generic [ref=e47]: Decoupled services · HTTP/3 QUIC
 - generic [ref=e48]: Encrypted data layer
 - generic [ref=e49]: UDP discovery
 - generic [ref=e51]: Web · Angular 19
 - generic [ref=e53]: Desktop · Tauri 2.0
 - generic [ref=e55]: Android · Kotlin
 - generic [ref=e57]: iOS · Swift
 - generic [ref=e59]: HelixTrack Core
 - generic [ref=e60]: Go + Gin · unified /do API
 - generic [ref=e62]: Auth service
 - generic [ref=e64]: Permissions service
 - generic [ref=e66]: Localization service
 - generic [ref=e68]: PostgreSQL / SQLite
 - generic [ref=e69]: SQLCipher AES-256
 - generic [ref=e71]: Redis cache
 - generic [ref=e81]: // architecture
- paragraph [ref=e82]:
 - strong [ref=e83]: JIRA alternative for the free world.
- heading "Summary" [level=2] [ref=e84]
- paragraph [ref=e85]: HelixTrack is a comprehensive, modern, open-source alternative to JIRA (and, via its Documents extension, Confluence) – a multi-platform project-management and issue-tracking system built on a Go microservices backend with native clients for web, desktop, and mobile.
- heading "Short description" [level=2] [ref=e86]

- paragraph [ref=e87]: Open-source JIRA/Confluence alternative. A Go microservices backend ("HelixTrack Core") exposes a unified REST API for project and issue tracking, plus a Confluence-style document workspace, served to native Web, Desktop, Android, and iOS clients over HTTP/3 QUIC.
- heading "Long description" [level=2] [ref=e88]
- paragraph [ref=e89]:
 - text: HelixTrack is an open-source project-management and issue-tracking platform positioned as a free-world alternative to JIRA and Confluence. Its heart is
 - strong [ref=e90]: HelixTrack Core
 - text: ", a REST API microservice written in Go with the Gin framework, offering full issue tracking, agile/scrum boards, team management, and a hierarchical permissions engine with swappable local or HTTP-backed implementations. Core exposes a distinctive single, action-routed"
 - code [ref=e91]: /do
 - text: endpoint with a consistent request/response envelope, and integrates with decoupled Authentication, Permissions, and Localization services that communicate over HTTP/3 QUIC and can run on separate machines or clusters. Data is stored in SQLite for development and PostgreSQL in production, encrypted at rest with SQLCipher (AES-256). A
 - strong [ref=e92]: Documents V2
 - text: "extension adds a Confluence-style workspace – spaces, pages, version control, templates, real-time WebSocket collaboration, and analytics. Around Core sit multiple client applications: an Angular web client, a Tauri + Angular desktop client, native Android (Kotlin) and iOS (Swift) apps, plus HarmonyOS and Aurora OS clients and a screensaver – all talking to the same backend and discoverable on local networks via UDP broadcast. The client applications are maintained as separate, private repositories and are presented here at the product level only."
- heading "Why we built it" [level=2] [ref=e93]
- paragraph [ref=e94]: To give teams a genuinely open, self-hostable replacement for the JIRA + Confluence stack – "for the free world" – without vendor lock-in, combining enterprise-grade tracking, documents, and collaboration under one open-source license.
- heading "Why it's a game-changer" [level=2] [ref=e95]
- paragraph [ref=e96]: It collapses two heavyweight commercial products (issue tracking + wiki/documents) into one open, high-performance, self-hostable platform with true multi-platform native clients. The HTTP/3-everywhere, fully-decoupled microservices design and at-rest SQLCipher encryption target performance and security levels usually reserved for proprietary SaaS, while remaining open and free to run anywhere.
- heading "What's innovative" [level=2] [ref=e97]
- list [ref=e98]:
 - listitem [ref=e99]:

- text: Unified action-based
- code [ref=e100]: /do
- text: API – one endpoint, action-routed, with a consistent envelope, instead of sprawling REST surface area.
- listitem [ref=e101]: HTTP/3 QUIC as the default inter-service transport.
- listitem [ref=e102]: Swappable permissions engine (local or HTTP implementation) and optional, independently-deployable services (Auth, Permissions, Localization).
- listitem [ref=e103]:
 - text: Multi-space data isolation via a
 - code [ref=e104]: "--space-root"
 - text: flag (per-project isolated databases + assets).
- listitem [ref=e105]: SQLCipher AES-256 encryption at rest.
- listitem [ref=e106]: Automatic client-to-server discovery via UDP broadcast on local networks.
- listitem [ref=e107]: Documents V2 "Confluence alternative" with optimistic-locking parallel editing and full change history.
- heading "Biggest technical challenges & how we solved them" [level=2] [ref=e108]
- list [ref=e109]:
 - listitem [ref=e110]:
 - strong [ref=e111]: Cross-platform native clients on one backend.
 - text: Solved by a strict single-backend contract (unified
 - code [ref=e112]: /do
 -
 - text: API + envelope) that every client – Web/Angular, Desktop/Tauri, Android/Kotlin, iOS/Swift, HarmonyOS, Aurora – targets identically, with UDP-broadcast service discovery to locate Core.
 - listitem [ref=e113]:
 - strong [ref=e114]: Low-latency, decoupled services.
 - text: Solved by adopting HTTP/3 QUIC across all inter-service calls and allowing each service (Auth, Permissions, Localization) to run independently and be disabled in test configs.
 - listitem [ref=e115]:
 - strong [ref=e116]: Confluence-grade collaborative documents.
 - text: Documents V2 adds spaces/pages/versioning with optimistic locking, conflict detection, full change history, and real-time WebSocket sync.
 - listitem [ref=e117]:
 - strong [ref=e118]: Encryption without killing performance.
 - text: SQLCipher AES-256 at rest plus multi-layer caching (in-memory LRU + Redis in the Localization service) to offset overhead.
- heading "Tech stack" [level=2] [ref=e119]

- list [ref=e120]:
 - listitem [ref=e121]:
 - strong [ref=e122]: Go + Gin
 - text: – chosen for high-throughput, low-latency HTTP services; implements Core's REST API, middleware (JWT/CORS), and the
 - code [ref=e123]: /do
 - text: router.
 - listitem [ref=e124]:
 - strong [ref=e125]: HTTP/3 QUIC
 - text: – used as the transport between Core and its Auth/Permissions/Localization services to cut latency and enable connection resilience.
 - listitem [ref=e126]:
 - strong [ref=e127]: PostgreSQL (prod) / SQLite (dev)
 - text: – relational store for the large tracking + documents schema; SQLite keeps local dev zero-setup, Postgres scales in production via a
 - code [ref=e128]: production
 - text: compose profile.
 - listitem [ref=e129]:
 - strong [ref=e130]: SQLCipher (AES-256)
 - text: – transparent database-level encryption at rest for sensitive project data.
 - listitem [ref=e131]:
 - strong [ref=e132]: Redis
 - text: – multi-layer caching in the Localization service (in-memory LRU + Redis) for fast multi-language lookups.
 - listitem [ref=e133]:
 - strong [ref=e134]: Uber Zap + Lumberjack
 - text: – structured, rotated logging in Core.
 - listitem [ref=e135]:
 - strong [ref=e136]: golang-jwt / JWT
 - text: – authentication and the
 - code [ref=e137]: jwt
 - text: field in the API envelope.
 - listitem [ref=e138]:
 - strong [ref=e139]: Angular 19 (+ Material, RxJS)
 - text: – the browser web client.
 - listitem [ref=e140]:
 - strong [ref=e141]: Tauri 2.0 + Rust + Angular
 - text: – the cross-platform desktop client (small footprint, native shell).
 - listitem [ref=e142]:
 - strong [ref=e143]: Kotlin (Android) / Swift + SwiftUI (iOS)
 - text: – native mobile clients.
 - listitem [ref=e144]:
 - strong [ref=e145]: Docker / Docker Compose (Podman-compatible)
 - text: – containerized deployment with
 - code [ref=e146]: /health

- text: checks.
- listitem [ref=e147]:
 - strong [ref=e148]: Testify (Go); Cypress/Playwright/Karma+Jasmine (clients)
 - text: – layered automated testing across backend and clients.
- heading "Status & honesty notes" [level=2] [ref=e149]
- list [ref=e150]:
 - listitem [ref=e151]:
 - strong [ref=e152]: "Status: beta."
 - text: HelixTrack Core is a working REST API microservice; the
 - strong [ref=e153]: Documents V2
 - text: extension is documented as approximately 95% complete with a known database field-mapping issue, so it is not presented as fully shipped.
 - listitem [ref=e154]:
 - strong [ref=e155]: "License: TBD."
 - code [ref=e156]: CLAUDE.md
 - text: states MIT, but the
 - code [ref=e157]: core/LICENSE
 - text: file of record is Apache 2.0 – this discrepancy must be resolved before the license is stated definitively.
 - listitem [ref=e158]: Performance figures cited in the project's README (e.g. 50,000+ requests/second, sub-millisecond query times) are stated design/marketing targets, not independently published benchmarks, and are therefore omitted from the claims above.
 - listitem [ref=e159]:
 - text: The client applications (Web, Desktop, Android, iOS, Aurora, HarmonyOS) live in
 - strong [ref=e160]: private
 - text: repositories and are described only at the product level.
- paragraph [ref=e161]:
 - strong [ref=e162]: "Priority tier:"
 - text: Helix-primary and the flagship of the Helix-Track product line – ranked before any Server Factory projects.
- contentinfo [ref=e163]: © 2026 Miloš Vasić – built on the OpenDesign system.

Test source

```

1 | const { test, expect } = require('@playwright/test');
2 | const { AxeBuilder } = require('@axe-core/playwright');
3 | const BASE = 'http://localhost:8082';
4 |
5 | test.describe('milosvasic.ru – accessibility', () => {
6 |

```

```

7 | // Coverage extended (not shrunk): axe-core now scans
  | the reframed homepage,
8 | // a representative product page, and the portfolio
  | index – the three surfaces
9 | // a visitor navigates. We disable only heading-order
  | (skills use h4 under h2
10 | // as sub-headings) and link-in-text-block (inline
  | accent-coloured links, a
11 | // common accessible pattern); every other WCAG rule,
  | including color-contrast,
12 | // is enforced.
13 | const PAGES = [
14 |   { name: 'homepage', url: BASE },
15 |   { name: 'product', url: `${BASE}/products/
  | helixtrack.html` },
16 |   { name: 'portfolio', url: `${BASE}/portfolio/` },
17 | ];
18 |
19 | for (const p of PAGES) {
20 |   test(`axe-core accessibility scan – ${p.name} (no
  | critical/serious violations)`, async ({ page }) => {
21 |     await page.goto(p.url);
22 |     const results = await new AxeBuilder({ page })
23 |       .disableRules(['heading-order', 'link-in-text-
  | block'])
24 |       .analyze();
25 |     const critical = results.violations.filter(v =>
  | v.impact === 'critical' || v.impact === 'serious');
26 |     // Surface the exact offending nodes so any failure
  | is precisely actionable.
27 |     if (critical.length > 0) {
28 |       console.log(`AXE ${p.name} critical/serious:`,
  | JSON.stringify(
29 |         critical.map(v => ({ id: v.id, impact:
  | v.impact, help: v.help,
30 |           nodes: v.nodes.slice(0, 6).map(n =>
  | ({ target: n.target, summary: n.failureSummary })),
  | null, 2));
31 |     }
32 |     if (results.violations.length > 0) {
33 |       console.log(`INFO ${p.name} non-critical:`,
  | JSON.stringify(results.violations.map(v => ({ id: v.id,
  | impact: v.impact, help: v.help })))));
34 |     }
> 35 |     expect(critical, `${p.name} must have no critical/
  | serious ally violations`).toEqual([]);
  |
  | ^ Error: product must have no critical/serious ally violations
36 |   });
37 | }
38 |
39 | test('heading hierarchy is reasonable (no deep skips)',
  | async ({ page }) => {
40 |   await page.goto(BASE);
41 |   const headingLevels = await page.evaluate(() => {

```

```

42 |     const els = document.querySelectorAll('h1, h2, h3,
    |     h4, h5, h6');
43 |     return Array.from(els).map((el) =>
    |     parseInt(el.tagName.substring(1)));
44 | });
45 | expect(headingLevels.length).toBeGreaterThan(0);
46 | // First heading should be h1
47 | expect(headingLevels[0]).toBe(1);
48 | // Only flag major skips (level jumps > 2), since h4
    | is used intentionally
49 | // as skill-group sub-heading under h2 section
    | headings.
50 | let prev = headingLevels[0];
51 | for (let i = 1; i < headingLevels.length; i++) {
52 |     const level = headingLevels[i];
53 |     if (level <= prev) { prev = level; continue; }
54 |     // Allow increase by up to 2 levels (h1→h3, h2→h4
    | are intentional)
55 |     if (level - prev > 2) {
56 |         // For h1→h4+ or h2→h5+ skips, check if this is
    | an edge case
57 |         expect(level - prev).toBeLessThanOrEqual(3); //
    | Allow h1→h4 in some layouts
58 |     }
59 |     prev = level;
60 | }
61 | });
62 |
63 | test('focus indicators are visible when tabbing', async
    | ({ page }) => {
64 |     await page.goto(BASE);
65 |
66 |     // First, verify the stylesheet contains a
    | visible :focus-visible rule
67 |     const hasFocusRule = await page.evaluate(() => {
68 |         for (const sheet of document.styleSheets) {
69 |             try {
70 |                 for (const rule of sheet.cssRules) {
71 |                     const css = rule.cssText || '';
72 |                     if (css.includes(':focus-visible') &&
    | (css.includes('outline') || css.includes('box-shadow') ||
    | css.includes('border'))) {
73 |                         return true;
74 |                     }
75 |                 }
76 |             } catch (e) { /* cross-origin */ }
77 |         }
78 |         return false;
79 |     });
80 |     expect(hasFocusRule).toBeTruthy();
81 |
82 |     // Now verify tabbing moves focus to interactive
    | elements.

```

```

83 | // In headless WebKit, Tab may not move focus at all
    | (known limitation).
84 | // The CSS rule check above is the source of truth;
    | tabbing is a secondary validation.
85 | let foundVisibleFocus = false;
86 | for (let i = 0; i < 20; i++) {
87 |     await page.keyboard.press('Tab');
88 |     await page.waitForTimeout(80);
89 |     const result = await page.evaluate(() => {
90 |         const el = document.activeElement;
91 |         if (!el || el === document.body) return { ok:
    | false, tag: 'body' };
92 |         return { ok: true, method: 'activeElement', tag:
    | el.tagName };
93 |     });
94 |     if (result.ok) { foundVisibleFocus = true; break; }
95 | }
96 |
97 | // Fallback: skip-link must exist with non-zero
    | outline width in CSS
98 | const skipLinkOk = await page.evaluate(() => {
99 |     const sl = document.querySelector('.skip-link');
100 |     if (!sl) return false;
101 |     const s = getComputedStyle(sl);
102 |     return parseFloat(s.outlineWidth) > 0 ||
    | s.outlineStyle !== 'none';
103 | });
104 |
105 | // Either tabbing works, or the skip-link has visible
    | focus styles in CSS
106 | expect(foundVisibleFocus || skipLinkOk).toBeTruthy();
107 | });
108 |
109 | test('prefers-reduced-motion is respected', async
    | ({ page }) => {
110 |     // Check that the stylesheet contains a @media
    | (prefers-reduced-motion: reduce) rule
111 |     await page.goto(BASE);
112 |     const hasReducedMotionRule = await page.evaluate(()
    | => {
113 |         for (const sheet of document.styleSheets) {
114 |             try {
115 |                 for (const rule of sheet.cssRules) {
116 |                     if (rule.media &&
    | rule.media.mediaText.includes('prefers-reduced-motion')) {
117 |                         return true;
118 |                     }
119 |                 }
120 |             } catch (e) {
121 |                 // Cross-origin stylesheets may throw – skip
    | them
122 |             }
123 |         }
124 |         return false;

```

```
125 |     });
126 |     expect(hasReducedMotionRule).toBeTruthy();
127 |
128 |     // Now emulate reduced motion and verify reveal
elements are immediately visible
129 |     await page.emulateMedia({ reducedMotion: 'reduce' });
130 |     await page.goto(BASE);
131 |     // With reduced motion, .reveal elements should
have .in class immediately
132 |     const revealCount = await
page.locator('.reveal').count();
133 |     const inCount = await
page.locator('.reveal.in').count();
134 |     expect(inCount).toBe(revealCount);
135 |     });
```